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```
# UBER PICKUPS ANALYSIS PROJECT
```

```
# January 2015 - June 2015
```

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#####
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```
#=====
```

```
# 1. LOAD LIBRARIES
```

```
#=====
```

```
library(tidyverse)
```

```
library(ggplot2)
```

```
library(dplyr)
```

```
library(janitor)
```

```
library(corrplot)
```

```
#=====
```

```
# 2. IMPORT DATA
```

```
#=====
```

```
Uber <- read.csv("C:/221222482/Uber Project/Uber.csv")
```

```
#=====
```

```
# 3. DATA CLEANING
```

```
#=====
```

```
# Check structure
```

```
str(Uber)
```

```
# Missing values
```

```
colSums(is.na(Uber))
```

```
# Duplicates
```

```
sum(duplicated(Uber))
```

```
# Replace missing borough values
```

```
Uber$borough[is.na(Uber$borough)] <- "Unknown"
```

```
# Convert date column
```

```
Uber$pickup_dt <- as.POSIXct(
```

```
  Uber$pickup_dt,
```

```
  format = "%Y-%m-%d %H:%M:%S"
```

```
)
```

```
#=====
```

```
# 4. FEATURE ENGINEERING
```

```
#=====
```

```
Uber$month <- format(Uber$pickup_dt,"%B")
```

```
Uber$hour <- format(Uber$pickup_dt,"%H")
```

```
Uber$day <- weekdays(Uber$pickup_dt)
```

```
Uber$day <- factor(
```

```
  Uber$day,
```

```
  levels = c(
```

```
    "Monday",
```

```
    "Tuesday",
```

```
    "Wednesday",
```

```
    "Thursday",
```

```
    "Friday",
```

```
    "Saturday",
```

```
    "Sunday"
```

```
)
```

```
)
```

```
# Seasons
```

```
Uber$season <- case_when(  
  Uber$month %in% c("December", "January", "February") ~ "Winter",  
  Uber$month %in% c("March", "April", "May") ~ "Spring",  
  Uber$month %in% c("June", "July", "August") ~ "Summer",  
  TRUE ~ "Other"  
)
```

```
# Weekday vs Weekend
```

```
Uber$day_type <- ifelse(  
  Uber$day %in% c("Saturday", "Sunday"),  
  "Weekend",  
  "Weekday"  
)
```

```
#=====
```

```
# 5. DATA SUMMARY
```

```
#=====
```

```
summary(Uber)
```

```
Uber %>%
```

```
  summarise(  
    Total_Pickups = sum(pickups),  
    Average_Pickups = mean(pickups)  
  )
```

```
#=====
```

``` # 6. SEASONAL ANALYSIS ```

```
#=====
```

```
season_summary <- Uber %>%
```

```
  group_by(season) %>%
```

```
  summarise(
```

```
    Total_Pickups = sum(pickups),
```

```
    Average_Pickups = mean(pickups)
```

```
  )
```

```
print(season_summary)
```

```
ggplot(season_summary,
```

```
  aes(x = season,
```

```
    y = Average_Pickups,
```

```
    fill = season)) +
```

```
geom_col() +
```

```
geom_text(
```

```
  aes(label = round(Average_Pickups,0)),
```

```
  vjust = -0.5
```

```
) +
```

```
labs(
```

```
  title = "Average Uber Pickups by Season",
```

```
  x = "Season",
```

```
  y = "Average Pickups"
```

```
)
```

```
#=====
```

``` # 7. TEMPERATURE ANALYSIS ```

```
#=====
```

```
cor(Uber$temp, Uber$pickups)
```

```
ggplot(Uber,  
  aes(x = temp,  
    y = pickups)) +  
  geom_point(alpha = 0.2) +  
  geom_smooth(method = "lm")
```

```
#=====
```

```
# 8. BOROUGH ANALYSIS
```

```
#=====
```

```
borough_summary <- Uber %>%  
  group_by(borough) %>%  
  summarise(  
    Total_Pickups = sum(pickups),  
    Average_Pickups = mean(pickups)  
  ) %>%  
  arrange(desc(Total_Pickups))
```

```
print(borough_summary)
```

```
ggplot(  
  borough_summary,  
  aes(  
    x = reorder(borough, Total_Pickups),  
    y = Total_Pickups  
  )  
  ) +  
  geom_col() +
```

```

coord_flip() +
labs(
  title = "Total Pickups by Borough",
  x = "Borough",
  y = "Total Pickups"
)

#=====

# 9. HOLIDAY ANALYSIS

#=====

holiday_summary <- Uber %>%
  group_by(hday) %>%
  summarise(
    Total_Pickups = sum(pickups),
    Average_Pickups = mean(pickups)
  )

print(holiday_summary)

ggplot(
  holiday_summary,
  aes(
    x = hday,
    y = Average_Pickups,
    fill = hday
  )
) +
  geom_col() +
  labs(
    title = "Holiday vs Non-Holiday Pickups",

```

```

    x = "Holiday",
    y = "Average Pickups"
  )

#=====

# 10. DAY OF WEEK ANALYSIS

#=====

daily_avg <- Uber %>%
  group_by(day) %>%
  summarise(
    Average_Pickups = mean(pickups)
  )

print(daily_avg)

ggplot(
  daily_avg,
  aes(
    x = day,
    y = Average_Pickups
  )
) +
  geom_col() +
  geom_text(
    aes(label = round(Average_Pickups,0)),
    vjust = -0.5
  ) +
  labs(
    title = "Average Pickups by Day of Week",
    x = "Day",

```

```

    y = "Average Pickups"
  )

#=====

# 11. BOROUGH + DAY ANALYSIS

#=====

borough_day <- Uber %>%
  group_by(borough, day) %>%
  summarise(
    Avg_Pickups = mean(pickups),
    .groups = "drop"
  )

ggplot(
  borough_day,
  aes(
    x = day,
    y = Avg_Pickups,
    color = borough,
    group = borough
  )
) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2) +
  labs(
    title = "Average Pickups by Day and Borough",
    x = "Day",
    y = "Average Pickups"
  )

```



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#=====
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```
# 12. WEEKDAY VS WEEKEND
```

```
#=====
```

```
hourly_comparison <- Uber %>%
```

```
  group_by(day_type, hour) %>%
```

```
  summarise(
```

```
    Avg_Pickups = mean(pickups),
```

```
    .groups = "drop"
```

```
  )
```

```
ggplot(
```

```
  hourly_comparison,
```

```
  aes(
```

```
    x = as.numeric(hour),
```

```
    y = Avg_Pickups,
```

```
    color = day_type
```

```
  )
```

```
) +
```

```
  geom_line(linewidth = 1.2) +
```

```
  geom_point() +
```

```
  labs(
```

```
    title = "Weekday vs Weekend Demand by Hour",
```

```
    x = "Hour",
```

```
    y = "Average Pickups"
```

```
  )
```

```
#=====
```

```
# 13. WEATHER CORRELATIONS
```

```
#=====
```

```
cor(Uber$temp, Uber$pickups)
cor(Uber$sd, Uber$pickups)
cor(Uber$pcp01, Uber$pickups)
cor(Uber$spd, Uber$pickups)
cor(Uber$vsb, Uber$pickups)
```

```
#=====
```

```
# 14. MARKET SHARE
```

```
#=====
```

```
market_share <- Uber %>%
  group_by(borough) %>%
  summarise(
    Total_Pickups = sum(pickups)
  ) %>%
  mutate(
    Share = round(
      Total_Pickups /
        sum(Total_Pickups) * 100,
      2
    )
  )
```

```
print(market_share)
```

```
#####
```

```
# END OF PROJECT
```

```
#####
```